

Part A: Theory Concepts (Student Performance Analysis)

1. Mean, Median, Mode

- Mean is the average of all values.
- Median is the middle value when data is sorted.
- Mode is the most frequent value.
- Mean is affected by extreme values (outliers).
- Median is more reliable for skewed data.

Example: Marks of students in a class.

2. Variance and Standard Deviation

- Variance measures how spread out the data is.
- Standard deviation is the square root of variance.
- High values indicate more variation.
- Helps understand consistency of data.
- Used in statistical analysis and risk measurement.

Example: Difference in marks among students.

3. Normal Distribution

- It is a bell-shaped curve.
- Data is symmetrically distributed.
- Mean = Median = Mode.
- Defined by mean and standard deviation.
- Common in real-life data.

Example: Heights of people.

4. Skewness and Kurtosis

- Skewness measures asymmetry of data.
- Positive skew has a long right tail.
- Negative skew has a long left tail.
- Kurtosis measures peakness of data.
- High kurtosis means sharper peak.

Example: Income distribution.

5. Probability

- Measures likelihood of an event.
- Value ranges from 0 to 1.
- Empirical probability is based on data.
- Theoretical probability is based on formulas.
- Widely used in prediction.

Example: Probability of getting heads in a coin toss.

6. Independent vs Dependent Events

- Independent events do not affect each other.
- Dependent events are related.
- Probability changes in dependent events.
- Important in statistical modeling.
- Used in machine learning.

Example: Dice roll vs card draw.

7. Bayes Theorem

- Used to update probability with new data.
- Based on prior and likelihood.
- Helps in decision making.
- Used in AI and ML.
- Useful in classification problems.

Example: Disease prediction using test results.

8. Eigenvalues and Eigenvectors

- Used in matrix operations.
- Represent direction of data.
- Important in PCA (dimensionality reduction).
- Helps simplify complex data.
- Used in machine learning.

Example: Image compression.